

**Input: structural
decomposition $D(A_P)$**

five MetaPatterns $G, O_{\leq}, T^*, T^{*rev}, L^*$ +
two refinements D^*, E^* + relational
extension B^{rel}

**Step 1 – Invariant
extraction**

for each structural component s ,
compute the invariant set I_s of A_P

Step 2 – MR derivation

map each invariant to an MR via the
Translate operator

Step 3 – Quotient

form MetaPattern $m(s, [I])$ per $\sim_s$
equivalence class

Step 4 – Aggregation

collect MetaPatterns over all structural
components

**Output: MetaPattern set
 $M(A_P)$**

closed under Translate (Theorem 1) ·
poly-time under a finite generating set
(Theorem 2)